Speech: slides 10–15

Part B — the division of Germany (Redding and Sturm 2008)

About 7 minutes 30 seconds. *[red brackets]* = what to point at or do; // = a short pause. If Part B is drawn, open with slides 1–2 (Part A speech, slides 1–7), then say: "Let me go to Part B."

Slide 10 — A fake border on the 7.5°E meridian finds nothing

about 1 min 30 s

> *[figure]* Part B: Redding and Sturm. After the division of Germany, the 20 West German cities within 75 kilometres of the new border — the **treated** group — lost population relative to the rest of the West, the control group. The figure is their design: the two groups share the same trend until 1949, and diverge after the division.
>
> Point (a) is a **placebo**: a falsification test. We apply the same difference-in-differences where there is no treatment, and it should find nothing.
>
> *[code]* We draw a fake border on the meridian 7.5 degrees East. `dmer` is each city's distance from the meridian in kilometres. `samp` is the West *without* the 20 real border cities: we drop them because otherwise their real decline would contaminate the placebo control group. Of the remaining 99 cities, `pl_t` are the **58** within 75 km of the meridian, mostly the Rhine–Ruhr; `pl_c` are the other **41**. `did_pl` is the difference-in-differences of annual growth, post-war minus pre-war, as in act 0.
>
> *[the two numbers on the left]* The result. // The placebo is **minus 0.064** percentage points per year, t equal to minus 0.28, p equal to 0.78. The actual border gives **minus 0.688**, t equal to minus 3.9 — minus 0.75 in Redding and Sturm's specification. The placebo is less than one tenth of the real effect, and statistically zero.

Slide 11 — The null placebo supports the market-access reading

about 1 min

> *[left card]* What would a large negative placebo have meant? Redding and Sturm's identification rests on **parallel trends**: without the division, border cities and the rest of the West would have grown alike. A large negative estimate along an arbitrary line would say that western regions were already on different paths, or hit by other shocks: uneven bombing, the decline of the Ruhr, where refugees settled. // Then the minus 0.75 could not be read as the causal effect of lost market access. Our placebo is zero, so the loss is specific to the real border.
>
> *[right card]* Optional: the same placebo in the model. We applied the same groups to the simulated division of act 2, `mod_pa`, and the model gives **plus 0.052**. Why positive? The meridian strip is the part of the West farthest from the border: its exposure to the East is 0.085, against 0.122 for the control. So it loses *less* market access than the

control, and grows relative to it. The size is about half of the model's own treatment effect, minus 0.112, with the opposite sign.

Slide 12 — Sealing the border deletes the eastern terms

about 1 min 15 s

[card 1] Point (b): exposure. The price index is a sum with one term per supplier. By the definition of the trade share, **each term is pi-n-i times the price index** to the one minus sigma: share times total. So the eastern terms add up to **s-E** times the total, where s-E is the share of the city's spending on the East.

[formula] Sealing the border sets phi to zero for eastern suppliers, so their terms disappear. At given wages, what remains is **one minus s-E** times the old sum. // So the price index rises by the factor one minus s-E, to the minus one over sigma minus one: the more a city bought from the East, the more its cost of living rises.

[card 2] The bound. Shares sum to one: home, plus other western cities, plus the East. So s-E equals one minus the home share minus the other western shares: it is at most **one minus pi-n-n**, with equality only if the city buys nothing from other western cities.

["Intuition" box] The intuition: a city that buys mostly at home cannot be very exposed. A city that buys a lot elsewhere can be — but only if that elsewhere is the East.

Slide 13 — Exposure to the East ranks the losers almost perfectly

about 1 min 15 s

[code] Now in the model, at the 1939 equilibrium. `sE` sums each western city's trade shares over the eastern columns; `pnn` is the home share; then we correlate both with the simulated population change. The bound holds in all 119 West cities. Mean exposure is **0.176** for the treated, **0.100** for the control.

[left panel, first row of the table] The correlation between exposure and the simulated change is **minus 0.998**. In the model the only shock is losing the East, so exposure ranks the losers almost perfectly; general equilibrium rescales the shock but keeps the order.

[rows 2 and 3] The home share is only an upper bound on exposure. Among the 20 treated cities it works, **0.85**: they are all close to the border, so buying outside means buying from the East. Across all 119 it drops to **0.21**, // because in the Rhineland a low home share means trading with other *western* cities, which the border does not touch.

[box "Link to act 3"] The link to act 3: small cities buy less at home, 0.37 against 0.54, so they are more exposed and lose about twice as much: minus 0.140 against minus 0.067 in the model, minus 1.12 against minus 0.37 in the data.

Slide 14 — The dials change the size of the effect, not the ranking

about 1 min 15 s

> *[first row of the table]* Point (c), optional. At baseline the model gives **minus 0.11** percentage points per year against **minus 0.75** in the data, and a correlation of only **0.08** across cities. The first number is about the **size** of the effect, the second about the **ranking** of cities.
>
> *[left card, "Size"]* For the size, the model could give a stronger response with more mobility, epsilon, more agglomeration, lambda, steeper distance decay or lower sigma. It cannot capture channels that are not market access: border cities losing their role as hubs or administrative centres, firms moving west, war damage and refugees.
>
> *[right card, "Ranking"]* For the ranking, it could accommodate unequal fundamentals and real transport networks, but not census noise and city-specific shocks: the spread of growth is 1.09 in the data against 0.07 in the model.
>
> *[highlighted row]* The highlighted row is act 5: epsilon 100 and lambda one third. // The effect goes from minus 0.11 to **minus 0.34**, but the correlation with the data stays at **0.079**. The row below is Redding and Sturm's own simulation, with epsilon equal to infinity. The parameters scale the response, like a volume knob; they do not change who loses.

Slide 15 — Two models agreeing, not a model fitting the data

about 1 min 15 s

> *[navy box]* Point (d). In act 5, epsilon 100 and lambda one third give **minus 0.342**, against **minus 0.362** in Redding and Sturm's simulation.
>
> *[first card]* Which assumption on agglomeration is implicit in their model? // Their model is Helpman: free entry makes the number of varieties proportional to population, so bigger cities offer more varieties and a lower price index. By Part A(d), that is an agglomeration force equivalent to a spillover of **one over sigma minus one**: one third at sigma equal to 4. And epsilon must be large because they assume perfect mobility, the limit of epsilon to infinity.
>
> *[second card]* Why no lambda in their paper? Because it is not a free parameter: monopolistic competition and sigma pin it down. Our model makes the same force an explicit dial.
>
> *[code, then third card]* The code recalibrates and re-solves the division with those parameters, `p3`, and correlates the result with the data and with their simulation. City by city, the correlation with Redding and Sturm is **0.99997**; with epsilon 300 and 1000 we get minus 0.354 and minus 0.359, converging to their minus 0.362.

[look at the Professor] So is this evidence about the data? No: it is evidence that the two models are **the same system**. Both under-predict the data, minus 0.36 against minus 0.75, and both have the same low correlation with city growth, about 0.08.

[hand over] My colleague will now present Part C.

Numbers to remember. Placebo -0.064 ($t = -0.28$, $p = 0.78$) vs actual -0.688 (RS -0.75); 58 vs 41 cities · model placebo $+0.052$; s^E 0.085 vs 0.122 · $\text{corr}(s^E, \Delta) = -0.998$; π_{nn}: 0.85 (20 treated), 0.21 (119) · s^E 0.176 vs 0.100 · baseline -0.112, corr 0.08; act 5 -0.342, corr 0.079; RS -0.362; data -0.746 · corr with RS 0.99997.
Words to get right. the 20 border cities are the *treated*, not the control · the placebo strip loses *less* access · each term = share × total · *one over sigma minus one* · evidence about the *models*, not the data.